國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

Lab 10

姓名：　廖彥瑄
學號：　10810030

Lab10-Ex1.
Complement heapify function
Code

Max heap:

```
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;

// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
    vector<int> arr;
    ifstream file(filename);

    if (!file) {
        cerr << "Error opening file: " << filename << endl;
        return arr;
    }

    string line;
    while (getline(file, line)) { // 持續讀取整行內容
        stringstream ss(line);    // 創建字符串流
        string value;
        while (getline(ss, value, ',')) { // 用逗號分隔值
            try {
                arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
            }
            catch (exception& e) {
                cerr << "Invalid number format in file: " << value << endl;
            }
        }
    }

    file.close();
    return arr;
```

```
}

class MaxHeap {
public:
    vector<int> heap; // 儲存 Max Heap 的元素

    // 建立 Max Heap
    void buildMaxHeap(vector<int>& arr) {  //建立 Max Heap
        heap = arr;

        for (int i = (heap.size() / 2) - 1; i >= 0; i--) { // 從最後一個非葉子節點
開始向上執行 Max Heap
            heapify(i);
        }
        sort(heap.begin(),heap.end(),greater<int>());//由大到小排列
    }

    void heapify(int i) {// 堆化函式（確保以 i 為根的子樹符合 Max Heap 性質）
        int largest = i;          // 假設當前節點是最大的
        int left = 2 * i + 1;     // 左子節點索引
        int right = 2 * i + 2;    // 右子節點索引

        if(left<heap.size()&&heap[left]>heap[largest]){//如果左子節點比當前節點大
            largest=left;
        }
        if(right<heap.size()&&heap[right]>heap[largest]){//如果看右子節點比當前節點
大
            largest=right;
        }
        if(largest!=i){//如果當前節點不是最大
            swap(heap[i],heap[largest]);//交換
            heapify(largest);
        }
    }

    // 顯示 Heap 的內容(使用 BFS)
    void printHeap() {
        for (int val : heap) {// 遍歷 Max Heap 中的每個元素
```

```
            cout << val << " ";// 輸出元素
        }
        cout << endl;
    }
};

int main() {
    // 從文件讀取輸入元素
    string filename = "input1.txt";//請貼上 input 檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信息
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建 Max Heap 對象
    maxHeap.buildMaxHeap(arr); // 建立 Max Heap

    // 輸出 Max Heap 的內容
    cout << "Max Heap: ";
    maxHeap.printHeap();
    cout << endl;

    system("pause");
    return 0;
}
```

Min heap:

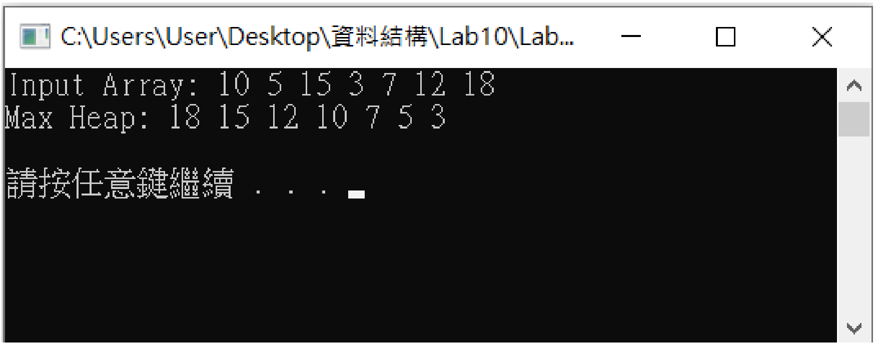
```
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
```

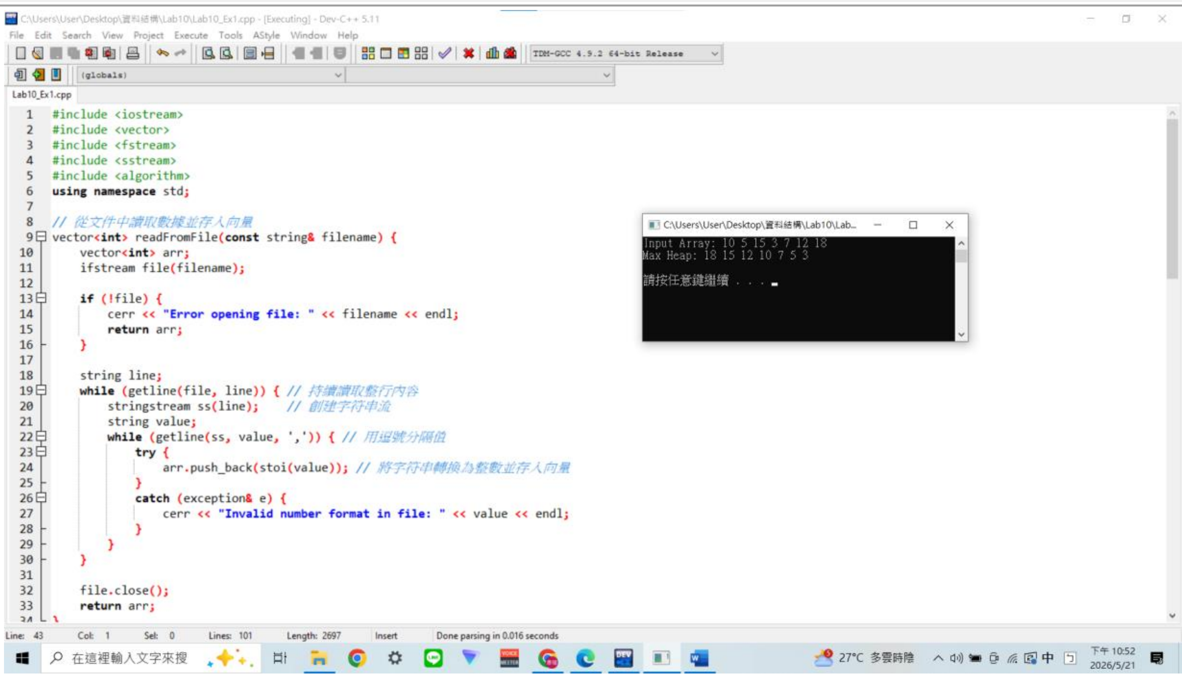
```cpp
#include <algorithm>
using namespace std;

// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
    vector<int> arr;
    ifstream file(filename);

    if (!file) {
        cerr << "Error opening file: " << filename << endl;
        return arr;
    }

    string line;
    while (getline(file, line)) { // 持續讀取整行內容
        stringstream ss(line);    // 創建字符串流
        string value;
        while (getline(ss, value, ',')) { // 用逗號分隔值
            try {
                arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
            }
            catch (exception& e) {
                cerr << "Invalid number format in file: " << value << endl;
            }
        }
    }

    file.close();
    return arr;
}

class MinHeap {
public:
    vector<int> heap; // 儲存 Min Heap 的元素

    // 建立 Min Heap
    void buildMinHeap(vector<int>& arr) {  //建立 Min Heap
        heap = arr;
```

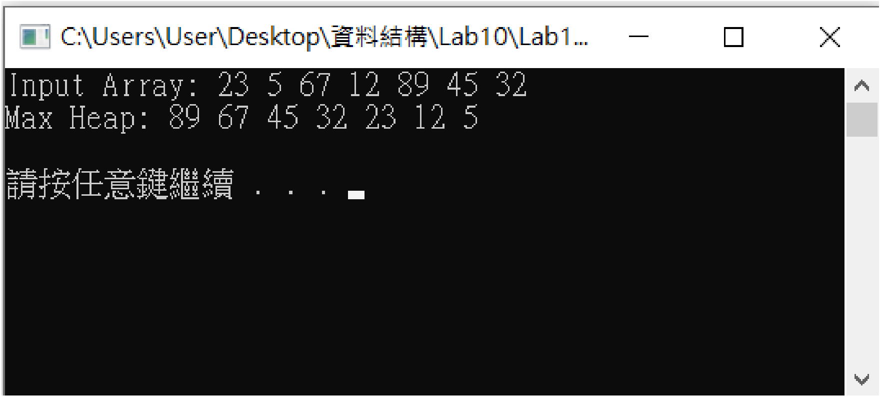
```
        for (int i = (heap.size() / 2) - 1; i >= 0; i--) { // 從最後一個非葉子節點開始向上執行Min Heap
            heapify(i);
        }
        sort(heap.begin(),heap.end());//由小到大排列
    }

    void heapify(int i) {// 堆化函式（確保以 i 為根的子樹符合Min Heap 性質）
        int smallest = i;          // 假設當前節點是最小的
        int left = 2 * i + 1;      // 左子節點索引
        int right = 2 * i + 2;     // 右子節點索引

        if(left<heap.size()&&heap[left]<heap[smallest]){//如果左子節點比當前節點小
            smallest=left;
        }
        if(right<heap.size()&&heap[right]<heap[smallest]){//如果看右子節點比當前節點小
            smallest=right;
        }
        if(smallest!=i){//如果當前節點不是最小
            swap(heap[i],heap[smallest]);//交換
            heapify(smallest);
        }
    }

    // 顯示Heap 的內容(使用BFS)
    void printHeap() {
        for (int val : heap) {// 遍歷 Min Heap 中的每個元素
            cout << val << " ";// 輸出元素
        }
        cout << endl;
    }
};

int main() {
    // 從文件讀取輸入元素
    string filename = "input3.txt";//請貼上input 檔案的正確路徑
```

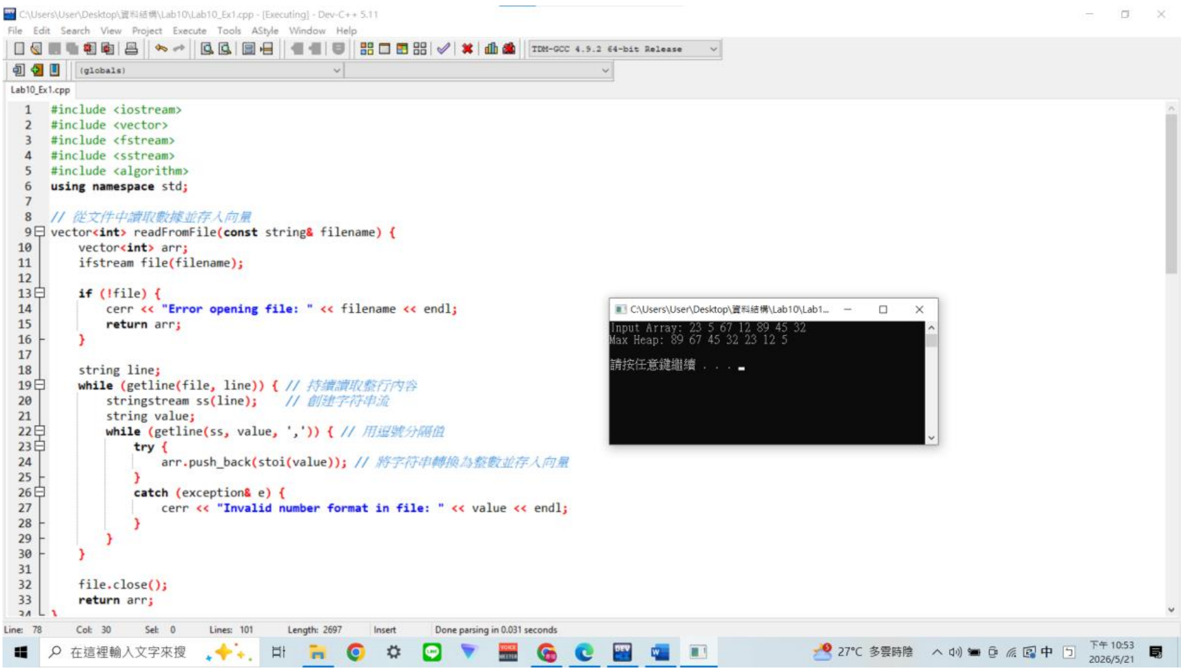
```
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信息
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MinHeap minHeap;// 創建Min Heap 對象
    minHeap.buildMinHeap(arr); // 建立Min Heap

    // 輸出Min Heap 的內容
    cout << "Min Heap: ";
    minHeap.printHeap();
    cout << endl;

    system("pause");
    return 0;
}
```

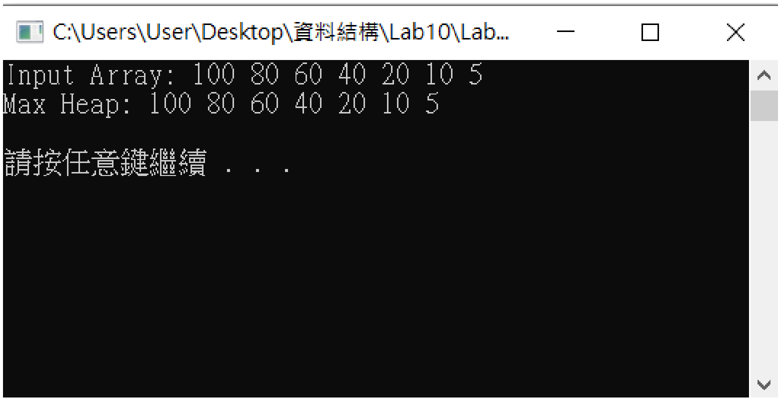
Result

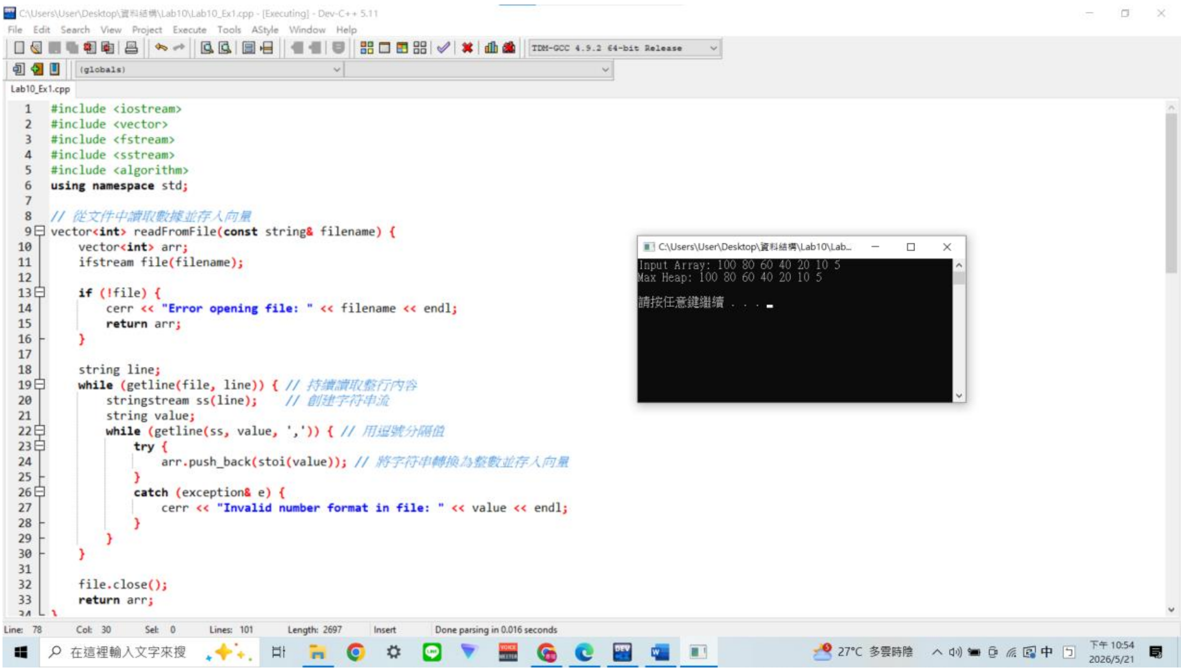


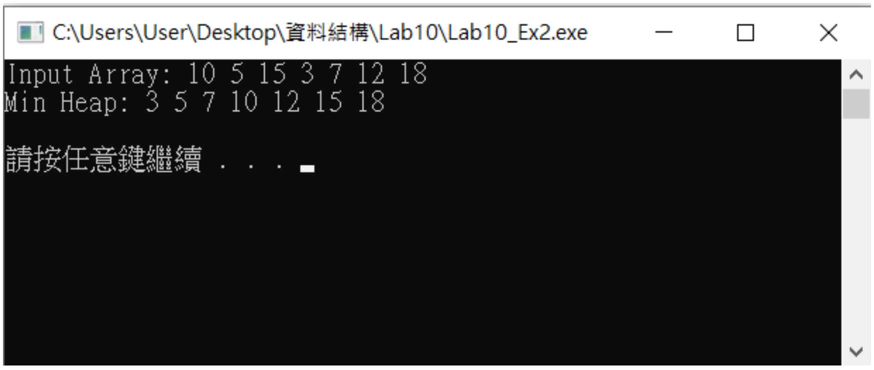

C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Lab10_Ex1.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
C:\Users\User\Desktop\資料結構\Lab10\Lab...
Input Array: 10 5 15 3 7 12 18
Max Heap: 18 15 12 10 7 5 3
請按任意鍵繼續 . . .
Line: 43 Col: 1 Sel: 0 Lines: 101 Length: 2697 Insert Done parsing in 0.016 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 10:52
2026/5/21

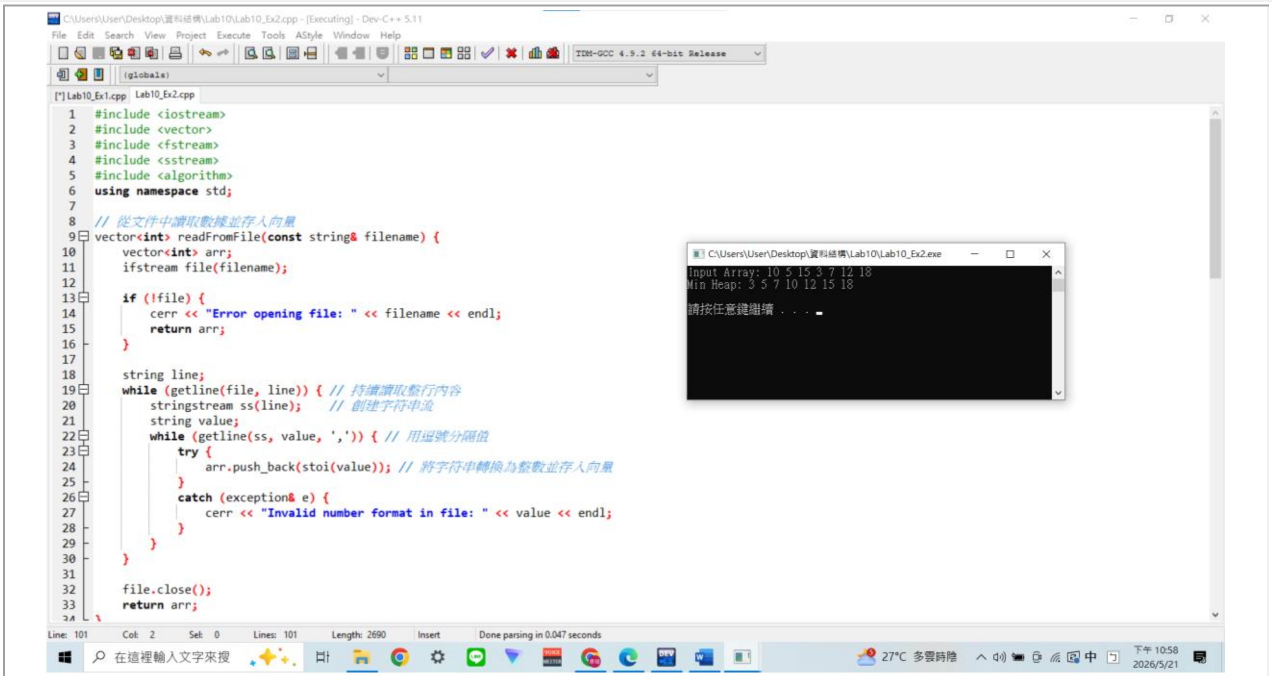

C:\Users\User\Desktop\資料結構\Lab10\Lab1...
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 67 45 32 23 12 5
請按任意鍵繼續 . . .

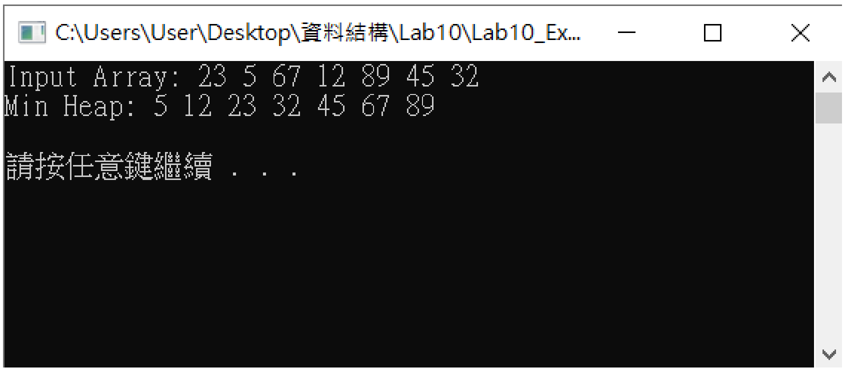

C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Lab10_Ex1.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
C:\Users\User\Desktop\資料結構\Lab10\Lab1...
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 67 45 32 23 12 5
請按任意鍵繼續 . . .
Line: 78 Col: 30 Sel: 0 Lines: 101 Length: 2697 Insert Done parsing in 0.031 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 10:53
2026/5/21

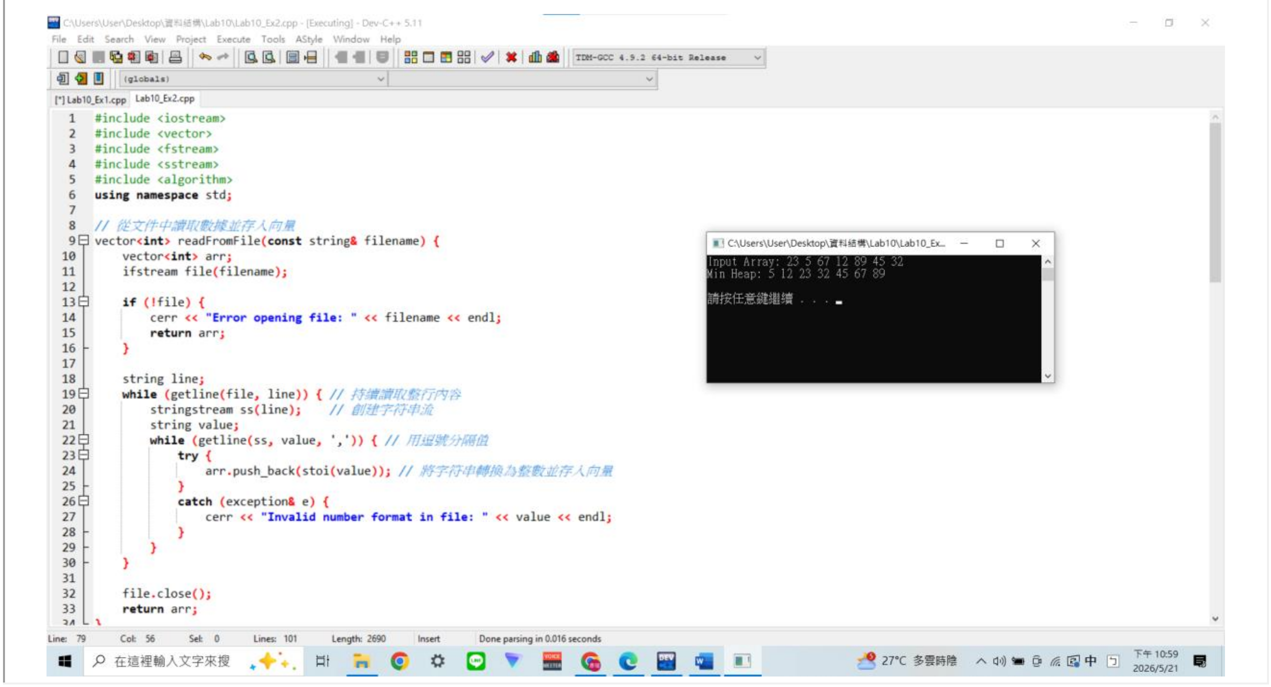

C:\Users\User\Desktop\資料結構\Lab10\Lab...
Input Array: 100 80 60 40 20 10 5
Max Heap: 100 80 60 40 20 10 5
請按任意鍵繼續 . . .

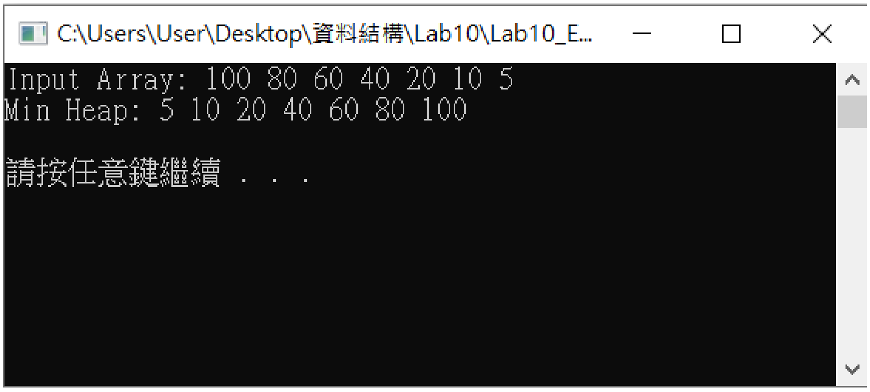

C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex1.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Lab10_Ex1.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
}
C:\Users\User\Desktop\資料結構\Lab10\Lab...
Input Array: 100 80 60 40 20 10 5
Max Heap: 100 80 60 40 20 10 5
請按任意鍵繼續 . . .
Line: 78 Col: 30 Sel: 0 Lines: 101 Length: 2697 Insert Done parsing in 0.016 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 10:54
2026/5/21

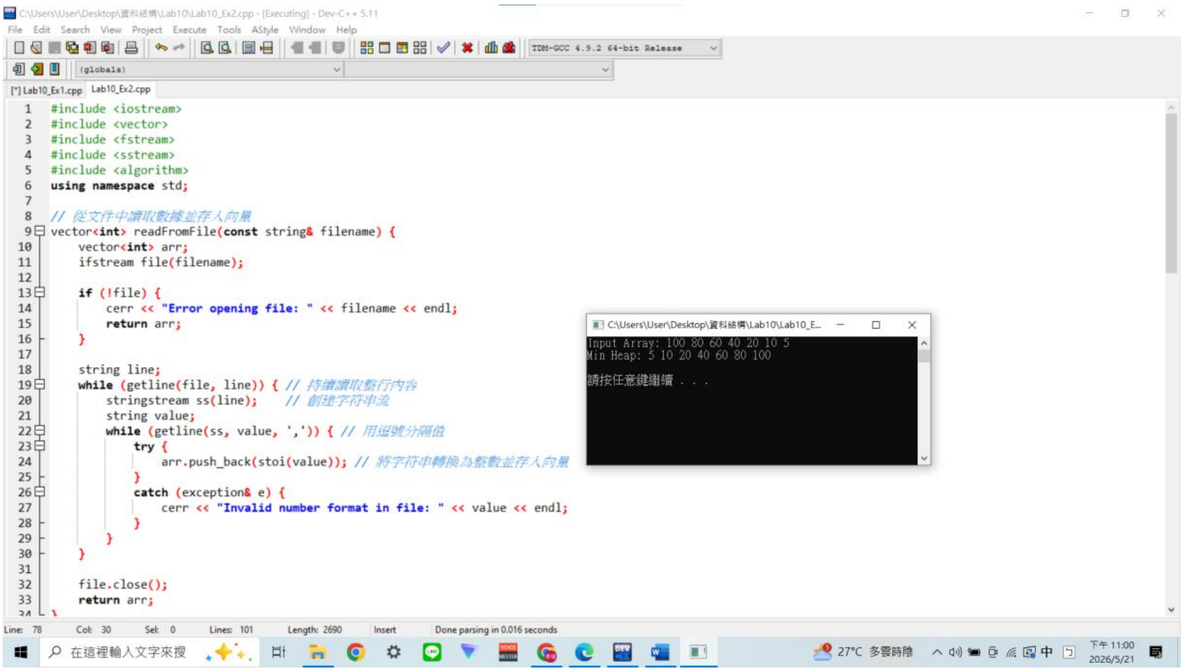

C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex2.exe
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 7 10 12 15 18
請按任意鍵繼續 . . .


C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex2.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
[*] Lab10_Ex1.cpp Lab10_Ex2.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex2.exe
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 7 10 12 15 18
請按任意鍵繼續 . . .
Line: 101 Col: 2 Sel: 0 Lines: 101 Length: 2690 Insert Done parsing in 0.047 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 10:58
2026/5/21


C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex...
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 23 32 45 67 89
請按任意鍵繼續 . . .


C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex2.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
[*] Lab10_Ex1.cpp Lab10_Ex2.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex...
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 23 32 45 67 89
請按任意鍵繼續 . . .
Line: 79 Col: 56 Sel: 0 Lines: 101 Length: 2690 Insert Done parsing in 0.016 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 10:59
2026/5/21

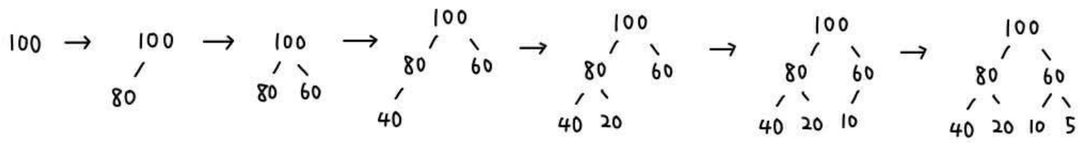

C:\Users\User\Desktop\資料結構\Lab10\Lab10_E...
Input Array: 100 80 60 40 20 10 5
Min Heap: 5 10 20 40 60 80 100
請按任意鍵繼續 . . .


C:\Users\User\Desktop\資料結構\Lab10\Lab10_Ex2.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
[*] Lab10_Ex1.cpp
Lab10_Ex2.cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
vector<int> arr;
ifstream file(filename);
if (!file) {
cerr << "Error opening file: " << filename << endl;
return arr;
}
string line;
while (getline(file, line)) { // 持續讀取整行內容
stringstream ss(line); // 創建字符串流
string value;
while (getline(ss, value, ',')) { // 用逗號分隔值
try {
arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
}
catch (exception& e) {
cerr << "Invalid number format in file: " << value << endl;
}
}
}
file.close();
return arr;
}
Input Array: 100 80 60 40 20 10 5
Min Heap: 5 10 20 40 60 80 100
請按任意鍵繼續 . . .
Line: 78 Col: 30 Sel: 0 Lines: 101 Length: 2690 Insert Done parsing in 0.016 seconds
在這裡輸入文字來搜
27°C 多雲時陰
下午 11:00
2026/5/21

Discussion

題目：100, 80, 60, 40, 20, 10, 5

Max heap：



Min heap：